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Chinese EFL learners' GenAI literacy in digital multimodal composing and self-regulated writing: chain mediation effects of needs satisfaction and creative self-concept

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ABSTRACT

Purpose: Digital multimodal composing (DMC) has emerged as a new literacy practice in English as a foreign language (EFL) writing that involves the production of a text comprising linguistic and non-linguistic meaning-making resources. It has been assumed that EFL learners with high levels of Generative Artificial Intelligence (GenAI) literacy in DMC would increasingly exercise their writing self-regulated learning (SRL) skills due to more opportunities for expression and activation of thoughts. However, the psychological mechanism underlying the impacts of students' GenAI literacy in DMC on their writing SRL remains underexplored, which is the research purpose of our study.

Methodology: Based on the self-determination theory and exploration and exploitation theory, this study developed a chain-mediation model to investigate whether needs satisfaction and creative self-concept can mediate the relationship between GenAI literacy in DMC and writing SRL. A total of 634 Chinese EFL learners were recruited to participate in this research and completed online questionnaires that measured GenAI literacy in DMC, needs satisfaction, creative self-concept, and self-regulated writing. And then the data analysis was performed through structural equation modeling in AMOS 28.0.

Findings and value: Our study found that creative self-concept exerted significant mediation effects between GenAI literacy in DMC and writing SRL, while needs satisfaction did not. And the chain mediation effects of needs satisfaction and creative self-concept were significant. Our research may contribute to developing a new syllabus that integrates GenAI literacy in DMC as a new literacy practice for promoting students' writing skills.

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Generative artificial intelligence; digital multimodal composing; self-regulated learning; needs satisfaction; creative self-concept

1. Introduction

The increasing integration of digital multimodal composing (DMC) into the writing pedagogy has been corroborated to be conducive to students' self-regulated learning (SRL) (Chiu 2024; Xia, Chiu, and Chai 2023a; Xia et al. 2023b). DMC refers to a type of creative

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practice that hinges on miscellaneous digital tools to generate texts through interweaving various semiotic multimodal resources, including images, written texts, spoken language, or sound, among other things (Abdelhalim 2024).

Recently, the emergence of generative artificial intelligence (GenAI) has offered new possibilities for DMC to enhance students' writing SRL, which refers to self-initiated behaviors that writers utilise to achieve multiple literary goals, such as enhancing their writing products and writing skills (Smith et al. 2025). In contrast to previous AI technologies that perform data prediction predicated on the past behaviors, GenAI tools integrate functionalities such as automated writing evaluation, visualization of goals or operational procedure, and machine translation techniques to produce new multimodal and textual outputs (Liu et al. 2024; 2025; Cosentino et al. 2025), which would significantly expand the DMC repertoire in the writing SRL classroom.

It should also be noted that writing SRL hinges on individual personal cognitive characteristics, including individual writing literacy or the motivation (a conative factor) that could be induced by the needs motivation. Furthermore, the interaction between humans and GenAI during DMC processes charts new ways for idea generation for more fluent writing SRL (Habib et al. 2024). Currently, the extant research mostly centers on intervention studies inspecting the effects of GenAI in DMC on EFL learners' writing SRL without further tapping into the underlying psychological mechanism (Liu et al. 2024; Kong and Yang 2024; Wang, Li, and Bonk 2024). It is imperative that EFL learners call upon their cognitive and conative resources to fully make use of the potential of GenAI technologies in their DMC practices to improve their writing SRL outcomes (Zhao, Lan, and Chen 2023). Considering the existing research gaps, our study aims to delve into the relationship between GenAI literacy in DMC and writing SRL and the chain mediation roles of needs satisfaction and creative self-concept that serve as mediators between these two factors.

2. Literature review

2.1. GenAI literacy in DMC and writing SRL

Recent years have witnessed significant developments in developing multimodal generative models such as DALL·E 3, Pika, and Sora, for writing SRL (Bird, Ungless, and Kasirzadeh 2023). These multimodal generative models were a double-edged sword. On the one hand, multimodal generative models endow EFL writers with a myriad of valuable opportunities to foster and develop their 'multimodal literacy,' or 'GenAI literacy in DMC,' which refers to the ability to generate, analyse, interpret, and access multimodal outputs that integrate various modes (e.g. text, images, audios, and videos) (Liu et al., 2025; Kang and Yi 2023). When EFL writers interact with multimodal generative models, it is imperative that they draw on their GenAI literacy in DMC through deciphering intended meanings, pondering the relationships among different modes of communication, and critically or ethically evaluating the appropriateness and effectiveness of multimodal works. On the other hand, the appearance of multimodal generative models has caused significant challenges for educators (Chang et al. 2023). These multimodal generative models may present concerns regarding students' potential unethical use of these tools, like engaging in acts of discrimination and exclusion, misuse, misinformation, and disinformation, etc. (Bird, Ungless, and Kasirzadeh 2023). Therefore, it is necessary for EFL learners to be

equipped with GenAI literacy in DMC to more effectively harness cutting-edge GenAI technologies and regulate their writing SRL processes.

Scholars have connected the use of or knowledge of multimodal GenAI to enhance writing SRL through an independent learning, vivid and indirect feedback mechanism, and strategic and purposeful decision-making procedures (Guo 2024; Hsu, Chang, and Jen 2024). GenAI literacy in DMC could impact how EFL learners integrate multimodal generative models as a kind of cognitive assistance in their writing SRL processes by eliciting critical thinking about the subject matter (Liu, Zhang, and Biebricher 2024). Moreover, a student's mastery level of self-regulation is fueled by his or her self-efficacy beliefs about successfully achieving desired outcomes. Therefore, GenAI knowledge in DMC could be regarded as a type of computer-related self-efficacy that exerts influences on students' intention and motivation to adopt GenAI for writing SRL (Xia, Chiu, and Chai 2023a; Xia et al. 2023b). Different lines of research have yielded incongruent findings regarding the influence of multimodal GenAI and writing SRL. For instance, Qin and other researchers (2024) investigated the mindset and usage patterns of writers during their interactions with a multimodal GenAI prototype system, CharacterMeet. Their study found that the GenAI system could be conducive to students' writing SRL processes by visualising the ideas and enhancing the immersion in the visual-and-audio environment. However, other researchers revealed the negative impacts of multimodal GenAI on students' writing SRL processes. For example, Chu, Lu, and Tu (2025) examined the undergraduates' attitudes towards incorporating multimodal GenAI to learn how to perform multimodal digital storytelling. Their study unveiled that it necessitates the students to regulate their reflection when they meditate on how to utilize the GenAI-generated sources to smooth their script writing. To overcome this obstacle, Chu, Lu, and Tu (2025) called for students to develop more literacy skills, that is, developing a higher level of GenAI literacy in DMC. Overall speaking, due to the inconsistent findings regarding the effects of multimodal GenAI on students' writing SRL process and the importance of multimodal GenAI literacy in this procedure, our study sets out to flesh out how GenAI literacy in DMC influences students' writing SRL.

2.2. Needs satisfaction as a mediator

Self-determination theory (SDT) is a widely utilized theory that describes humans' basic psychological needs to experience autonomy, competence, and relatedness (Ryan 2017). Autonomy denotes people's independent decision-making over their own actions. Competence refers to individuals' perceptions of their capability to finish the tasks. Relatedness represents people's experience of connecting with other people. To date, the extant studies have proven that the GenAI literacy in DMC may impact students' autonomy needs. For instance, by probing seven university students' photo essay construction processes with the help of GenAI, Tan, Xu, and Wang (2025) found that students' insufficient multimodal GenAI literacy limited their autonomous orchestration of both non-linguistic and linguistic resources to establish their identities in writing. And Jiang (2024)'s study revealed that harnessing critical multimodal GenAI literacy may be beneficial for students' autonomous writing SRL workflow.

Additionally, students with a high level of GenAI literacy in DMC can gain more competence through obtaining more informative multimodal feedback. Some researchers

have revealed that the quality of GenAI feedback that satisfies EFL learners' competence needs is affected by students' utilization of GenAI knowledge, such as prompt engineering skills (Woo, Guo, and Susanto 2024). Furthermore, students with a high level of GenAI literacy in DMC could get more opportunities to have human-like interaction opportunities attributed to many anthropomorphic features embedded in multimodal generative models, which may satisfy their need for relatedness (Zhu et al. 2025). Moreover, researchers have congruously speculated that as learners' needs are satisfied, they are willing to take greater responsibility for their writing, which necessitates them to develop and apply more writing SRL strategies (Han, Zhao, and Ng 2021; Mickwitz and Suojala 2020). Nevertheless, most of the studies that have examined the relationship between needs satisfaction and writing SRL in the online learning or language learning context have not tapped into the EFL writing context (Xu and Zhou 2024; Zhou et al. 2025). Their findings illustrated that satisfying students' needs was positively linked with some writing SRL strategies in terms of metacognitive, social regulation, and resource management dimensions. It is worth taking a deeper look into the degree to which these positive effects are generalisable to the GenAI-assisted EFL writing context. To fill the lacuna, our study attempts to look into the mediation role of needs satisfaction between GenAI literacy in DMC and writing SRL.

2.3. Creative self-concept as a mediator

Creative self-concept may play a mediation role between GenAI literacy in DMC and writing SRL. Creative self-concept is an amalgamation of novel ideas, meaning, or values to different stakeholders (Habib et al. 2024). Prior studies mostly focused on how writing SRL influenced creative self-concept without examining the reverse causal relationships (Urban and Urban 2025; Zielińska et al. 2025). Certain professional skills and knowledge are necessary for the emergence of new ideas since the originality of generated ideas is dependent on the degree to which people engage in delving deeply into their knowledge (Rietzschel, Nijstad, and Stroebe 2007). Social cognitive views of SRL regard learners as creative agents influenced by environmental variables such as social environment and peer feedback (Winne and Hadwin 2010). Chinese EFL learners highly literate in GenAI for DMC could benefit more from harmonious peer-learning and GenAI-human social interaction relationship (Fathi, Yousefi, and Sedighravesheh 2017), which could be deemed as a catalyst to promote students' creative mind-mapping through peer feedback for their writing SRL.

Furthermore, how creative self-concept impacts writing SRL in the GenAI context has also been underexplored. Underpinned by a theoretical framework of exploration and exploitation (Nielsen, Mathiassen, and Hansen 2018), individuals have to tap into their existing expertise while also delving into innovative ways of doing things to sustain their competitive edge. Thereby, it could be assumed that an individual's prospective success in writing SRL is a result of utilizing prior knowledge-GenAI literacy in DMC, and students' engagement in creative ways of doing things. EFL learners with creative thinking induced through interaction with multimodal generative models may use divergent thinking to facilitate idea generation in the planning stage during writing SRL (Hubert, Awa, and Zabelina 2024). In addition, the creative mindset could also help EFL learners adapt to the written situation and regulate emotions for monitoring in the writing SRL process. In the self-reflection

stage of writing SRL, EFL learners with creative thinking could be more open to others' ideas to integrate others' suggestions into their written texts (Rubenstein et al. 2020). Considering all the above research gaps and possible links between these variables undergirded by the exploration and exploitation theory, our study sets out to examine how creative self-concept acts as a mediator between GenAI literacy and writing SRL.

2.4. The chain mediation effect of needs satisfaction and creative self-concept

Needs satisfaction and creative self-concept may play a chain mediation role between GenAI literacy in DMC and writing SRL. To begin with, since the need for autonomy is positively correlated with the growth mindset, referring to the belief that an individual's basic attributes are malleable and can be cultivated through endeavors (Dweck 2015; Dweck, Chiu, and Hong 1995), encouraging individuals to act independently and corroborating their perspectives through GenAI-supported DMC activities may arouse a sense of control and mastery. This perspective may direct EFL learners to believe that basic creative self-concept attributes are malleable and writing SRL skills that can be cultivated through persistent efforts on creative self-concept, goal-planning, and idea generation (Qiu and Lee 2020). Second, certain professional skills and knowledge are necessary for the emergence of new ideas and methods since the originality of generated ideas is dependent on the degree to which people engage in deep exploration of their knowledge (Rietzschel, Nijstad, and Stroebe 2007). If the GenAI learners have a high level of GenAI literacy in DMC, they may more readily acquire the necessary knowledge and skills to improve their creative self-concept. The satisfaction of GenAI literacy needs may enhance their confidence in themselves, stimulate their enthusiasm to find and solve problems, prompt them to tackle more creative and demanding writing tasks, and provide the necessary energy for creative self-concept and writing SRL.

Third, the satisfaction of EFL learners' relatedness needs alleviates their concern about interpersonal loss and risk. Sociocognitive views of SRL regard learners as agents influenced by environmental variables (e.g. social modeling and feedback) (Winne and Hadwin 2010). Chinese EFL learners highly literate in GenAI for DMC could benefit more from harmonious peer-learning and GenAI-human interaction relationship, which could be deemed as a catalyst to promote students' creative mind-mapping through peer feedback for writing SRL (Fathi, Yousefi, and Sedighravesht 2017). Previous scholars have argued that there is a close relationship between needs satisfaction and creative self-concept, positing that when the basic psychological needs (i.e. autonomy, competence, and relatedness) are satisfied, intrinsic motivation will emerge, therefore leading to one's creative self-concept (Nong, Wei, and Li 2024; Urban, Pesout, and Urban 2024). Nevertheless, Enko's (2014) research disclosed that fulfilling needs satisfaction is a necessary, yet not a sufficient, precondition for creative self-concept. These contradictory findings warrant more studies to clarify the correlation between needs satisfaction and creative self-concept and how these two variables have chain mediation effects between GenAI literacy in DMC and writing SRL.

2.5. Formulation of research hypotheses from convergent theoretical foundations

Based on the literature review, we have found some theoretical convergences between SDT, the Exploration-Exploitation Theory (EET), and the social-cognitive model of SRL that could help formulate our research hypotheses.

Firstly, the self-determination theory (SDT) overlapped with Zimmerman's social-cognitive model of SRL. Both of these theoretical models highlight the importance of autonomous self-regulation. Self-determination theory reckons that intrinsic motivation and autonomous self-regulation are the desired motivation type (Orsini, Evans, and Jerez 2015). In social cognitive models, SRL also stresses students' autonomous self-regulation by claiming that it is imperative that students have active management of their own motivations and proactively engage with their own learning process (Zimmerman 2000). According to SDT, basic psychological needs theory centers on the psychological needs for autonomy, competence, and relatedness as the basis of students' autonomous self-regulation. The regulation of behavior when people's interests and values are the reason for acting is said to be autonomous. How students self-determine through regulating intrinsic motivation could result in more motives with an aim to generating autonomous self-regulation value and feelings (Gurtner et al. 2012; Niemiec and Ryan 2009).

Furthermore, both SRL and SDT stress that social and contextual factors affect individuals' learning and SRL motivation (Held and Mejeh 2024). While making meaning in DMC, learners remix visual, linguistic, gestural, aural, and spatial modes, offering a context for socially embedded literacy practices and students' identity presentation (Jiang and Hafner 2024). The 'multimodal remixing' characteristic embedded in the GenAI literacy in DMC occurs through the multiple modes' involvement in DMC (Xie and Jiang 2025), which expands the styles and ingredients for meaning-making repertoires, creating autonomous expression spaces to impact changes in learners' writing autonomy in students' needs satisfaction beliefs as well as their subsequent writing SRL behaviors. These theoretical overlapping between SRL and SDT justified the relationship between GenAI literacy in DMC and writing SRL (see H1) as well as pointing out the underlying mediation role of needs satisfaction between GenAI literacy in DMC and writing SRL (H2, 3, 4).

Secondly, there is alignment with EET and the social-cognitive model of SRL. According to exploration-exploitation theory, learners need to brainstorm for idea generation, which is also an essential component during writing SRL (Tran and Ma 2025). When EFL learners pursue a creative goal through exploring the new writing resources based on students' GenAI literacy in DMC and exploit their prior knowledge, they have to undergo the process of sustaining, revising, maintaining, and restrategizing (Ivcevic and Nusbaum 2017). This inherently unstable process is replete with uncertain challenges, and requires students' cognitive, motivational, and emotional engagement to adapt their efforts to cope with these challenges. This adaptive regulating process is akin to the SRL writing process which requires students to keep on adjusting and regulating their strategies (Zhou and Thomas 2025). As a result, creativity may have significant mediation effects on GenAI literacy in DMC and students' writing SRL (H5,6,7).

Last but not least, there is convergence between SDT and EET. SDT and EET converge in their emphasis on how individuals should navigate between novelty-seeking and goal-directed creative actions, though from different angles. SDT posits that the satisfaction of the three basic psychological needs of autonomy, competence, and relatedness results in enhanced intrinsic motivation. In terms of the exploration, the students who have intrinsic motivation could be more likely to explore their own strengths as writers, seek novelties that are enjoyable and conform to their interests, as humans are intrinsically motivated to explore and learn (Lin and Wang 2021; Meulenbroeks, van Rijn, and Reijerkerk 2024). With respect to exploitation, exploitation is creating reliability in experience,

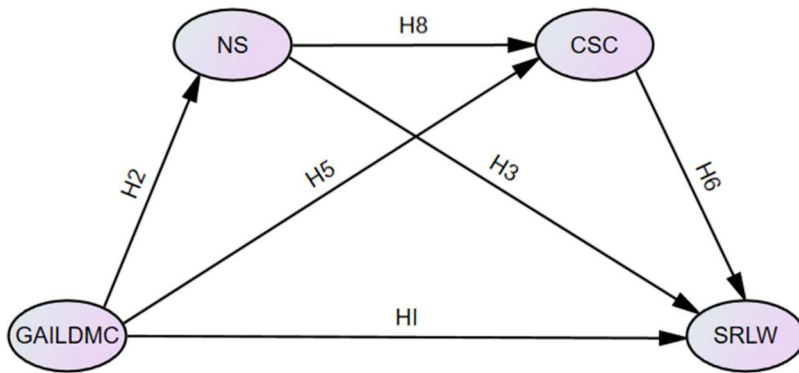


Figure 1. The proposed theoretical model. Note. GAILDMC: GenAI literacy in Digital Multimodal Composing; NS: Needs Satisfaction; CSC: Creative Self-Concept; SRLW: Self-regulated Learning of Writing.

focusing on the use, refinement, and reuse of the extant knowledge (Kane and Alavi 2007). Hence, the concept of exploitation overlaps with competence in SDT theory, as students with high competence in GenAI use in DMC had a more extensive existing knowledge base to leverage. Therefore, needs satisfaction may co-work with creative self-concept to act as the mediators between the GenAI literacy and writing SRL (H8, 9).

All hypotheses based on the above theoretical convergences were illustrated in Figure 1 except H4, H7, and H9. H4 can be supported if both H2 and H3 are correct. H7 can be corroborated if both H5 and H6 are valid. H9 can be verified if H2, H6, and H8 are all validated. The proposed hypotheses are demonstrated below:

H1: GenAI literacy in DMC is significantly positively correlated with writing SRL.

H2: GenAI literacy in DMC is significantly positively correlated with needs satisfaction.

H3: Needs satisfaction is significantly positively correlated with writing SRL.

H4: Needs satisfaction significantly mediates the association between GenAI literacy in DMC and writing SRL.

H5: GenAI literacy in DMC is significantly positively correlated with creative self-concept.

H6: Creative self-concept is significantly positively correlated with writing SRL.

H7: Creative self-concept significantly mediates the association between GenAI literacy in DMC and writing SRL.

H8: Needs satisfaction is significantly positively correlated with creative self-concept.

H9: GenAI literacy in DMC was found to indirectly predict writing SRL through a serial mediation pathway involving needs satisfaction and creative self-concept.

In order to examine these hypotheses, we developed four research questions to lead our investigation:

Research Question 1. To what degree does GenAI literacy in DMC predict writing SRL? (Addressing Hypothesis 1)

Research Question 2. To what extent does needs satisfaction mediate the relationship between GenAI literacy in DMC and writing SRL? (Addressing Hypotheses 2,3, 4)

Research Question 3. To what degree does creative self-concept mediate the relationship between GenAI literacy in DMC and writing SRL? (Addressing Hypotheses 5,6,7)

Research Question 4. To what extent do creative self-concept and needs satisfaction serially mediate the relationship between GenAI literacy in DMC and writing SRL? (Addressing Hypotheses 8, 9)

3. Materials and methods

3.1. Sampling and data collection

Prior to the distribution of the survey, ethical approval was granted from the ethics committee of the researchers' University (protocol number: UAHPEC28436). Drawing on purposive sampling techniques, we recruited questionnaire respondents by posting the questionnaire link to public GenAI discussion groups on different Chinese social media platforms such as Weibo and WeChat. Individuals who considered themselves Chinese EFL learners and had experience using GenAI technologies for DMC tasks were eligible to participate in this study. In terms of the use cases of EFL learners for GenAI-assisted DMC, some text-to-image platforms such as DALL-E 2 could elicit students' iterative revision and creation of improved ideas in the SRL reflecting stage (Liu et al. 2024). Moreover, GenAI-generated videos can facilitate idea creation during the SRL planning phase. For instance, the background and narrative elements embedded in the GenAI-driven text-to-video platform Fliki could elicit students' brainstorming of ideas for their descriptive writings (Yaojun and Tekwa 2025). Furthermore, the text-to-video application, Sora, can exercise students' critical thinking skills through creating concept videos in the SRL performance phase (Li, Xia, and Guo 2025).

Participants were required to read the introductory paragraph at the beginning of the online questionnaire to comprehend the research aim and to offer informed consent. All the questionnaires used in our study were anonymous to ensure the confidentiality of the data and privacy of the students. Students were invited to participate in the research on a voluntary basis. Structural Equation Modeling (SEM) suggested that the sample size should be at least 10 times the total amount of observed factors (Al Mamun et al. 2025). In line with this recommendation, a total of 850 Chinese EFL learners, aged 18–38 years old, were drawn from 16 schools across multiple regions in China and overseas, resulting in a response rate of 74.6% and the collection of 634 valid questionnaires.

3.2. Participant characteristics

The sample consisted of 634 participants with diverse sociodemographic characteristics. There were 57.6% males ($n = 365$) and 42.4% females ($n = 269$). The age distribution was as follows: participants aged 18–20 years accounted for 15.9% ($n = 101$), whereas 21–23 years made up the largest group at 30.8% ($n = 195$), followed by 24–26 years at 24.8% ($n = 157$). Relatively smaller proportions of participants were aged 27–29 years (14.7%, $n = 93$), 30–32 years (5.7%, $n = 36$), 33–35 years (6%, $n = 38$), and 36–38 years (1.9%, $n = 12$). The mean age of participants was 24.67 years ($SD = 4.41$). And their age ranged from 18 to 36 years old.

With regard to grade level, the majority were undergraduates at 58.5% ($n = 371$), followed by master's students at 18.5% ($n = 117$) and PhD students at 23% ($n = 146$). With respect to the school location, the participants were spread across various regions, with the largest groups from overseas (22.9%, $n = 145$), South China (19.6%, $n = 124$), and East China (18.9%, $n = 120$), followed by North China (14%, $n = 89$). Other regions included Central China (11%, $n = 70$), Southwest China (5%, $n = 32$), and Northwest China (2.7%, $n = 17$). A small number of participants were from Northeast China (5.5%, $n = 35$), Hong Kong, Macau, and Taiwan (0.3%, $n = 2$). The major types of participants were also diverse, with the highest proportion studying Science (34.5%, $n = 219$), followed by Humanities and Social Sciences (29.3%, $n = 186$), Engineering (23.5%, $n = 149$), Medicine (4.4%, $n = 27$), Agriculture (3%, $n = 19$), and Others (5.4%, $n = 34$).

3.3. Research instruments

3.3.1. GenAI literacy in digital multimodal composing scale

The Chinese EFL learners' GenAI literacy in DMC was assessed by the GenAI literacy in digital multimodal composing scale (GAILDMC), designed by Liu et al. (2025). It consists of 17 items, involving affective literacy (e.g. I use GenAI because it can build personalized experiences of digital multimodal composing.), behavioral literacy (e.g. I actively participate in GenAI-related learning activities for digital multimodal composing.), cognitive literacy (e.g. I know prompt engineering for GenAI in digital multimodal composing.), and ethical literacy (e.g. I am alert to discrimination and exclusion issues when using GenAI applications for digital multimodal composing.). This scale is formatted on a six-point Likert scale, from 'strongly disagree' to 'strongly agree'. The above 17-item four-factor structure of Chinese EFL students' GAILDMC was cross-validated in the present study with a sample of 634 Chinese EFL learners, suggesting an overall acceptable model fit ($\chi^2/df = 2.625$, GFI = 0.966, PGFI = 0.948, CFI = 0.971, PCFI = 0.736, RMSEA = 0.051). A scale analysis of the total 17 items in S revealed high internal consistency (Cronbach's $\alpha = 0.899$).

3.3.2. Self-regulated learning of writing scale

The EFL learners' self-regulated learning of writing was measured by the Writing Strategies for Self-Regulated Learning Questionnaire (WSSRLQ), developed by Teng and Zhang (2016). It is comprised of 40 items, covering cognitive strategies (e.g. 'When writing, I check the structure for logical coherence.'), metacognitive strategies (e.g. 'Before writing, I think about the core elements of a good composition I have learned to help me plan.'), social behavior strategies (e.g. 'I am open to peer feedback on my writing.'), and motivational regulation strategies (e.g. 'I find ways to regulate my mood when I want to give up writing.'). The scale uses a 6-point rating, with scores ranging from 1 to 6, ranging from 'strongly disagree' to 'strongly agree'. In this study, Cronbach's α was 0.845, which demonstrates relatively good reliability. Confirmatory factor analysis (CFA) results corroborated the good construct validity of the self-regulated learning scale within this study ($\chi^2/df = 2.763$, GFI = 0.978, PGFI = 0.521, CFI = 0.975, PCFI = 0.650, RMSEA = 0.053).

3.3.3. Needs satisfaction scale

Needs satisfaction was evaluated by the needs satisfaction scale established by Xia, Chiu, and Chai (2023a) and Xia et al. (2023b). It comprises three subscales of perceived

autonomy and is adapted to fit in the GenAI-assisted DMC context (e.g. 'I have many opportunities with the multimodal generative models to decide for myself how to learn.'). perceived competence ('I have been able to learn interesting new knowledge with the multimodal generative models.'). and perceived relatedness (When I learn with multimodal generative models, I feel comfortable.). The scale consists of 12 items, using a 6-point rating, with scores ranging from 1 to 6, representing 'strongly disagree' to 'strongly agree.' In this study, Cronbach's α was 0.878, which signifies relatively good reliability. The relatively satisfying construct validity is reflected in CFA fit indices ($\chi^2/df = 3.585$, GFI = 0.971, PGFI = 0.518, CFI = 0.971, PCFI = 0.647, RMSEA = 0.064).

3.3.4. Creative self-concept scale

Creative self-concept was measured by merging and adapting the *Short Scale of Creative Self* (Karwowski 2015) and the creative self-efficacy scale (Abulela 2024). The creative self-concept scale is composed of 13 items utilising the six-point Likert scale, measuring creative ideas self-efficacy (e.g. 'I am willing to think creatively in case of challenging situations.'). creative performance self-efficacy (e.g. 'I trust myself to perform creatively in different situations.'). and creative personal identity (e.g. 'Being a creative person is important to me.'). In this study, Cronbach's α of the creative self-concept scale was 0.869, representing excellent reliability. The relatively reasonable construct validity is reflected in the CFA fit indices ($\chi^2/df = 4.104$, GFI = 0.968, PGFI = 0.516, CFI = 0.963, PCFI = 0.642, RMSEA = 0.070).

3.4. Data analysis

Data analysis was carried out by using Amos 28.0 and SPSS 28.0. Initially, to filter out the potential negative impacts caused by common method bias, the Harman single-factor test was conducted. Afterwards, the sample's characteristics were interpreted through descriptive analysis. Thereafter, a CFA analysis was implemented to test the measurement model to inspect its construct validity. Subsequently, SEM was conducted to evaluate the structural model based on goodness-of-fit indices.

4. Results

4.1. Common method variance

As the results of self-reported variables could be affected by common method bias, Harman's (1976) single-factor test in SPSS 28.0 was performed. All the measurement items of the four variables (i.e. GenAI literacy in DMC, SRL in writing, needs satisfaction, and creative self-concept) were examined through exploratory factor analysis. According to the test result, the first factor comprised 39.39% of the total variance, which was less than the critical benchmark of 40% (Zhang, Su, and Wu 2023). These findings substantiate that there is no common method bias in this study, signifying that the study's reliability and validity were not substantially affected.

4.2. Measurement model

In this study, the skewness of all items in each scale ranged from -0.565 to -0.098 , and the kurtosis ranged from -0.897 to 0.877 , which indicated that the kurtosis and

skewness values of all items were between -2 and $+2$, suggesting the data were in normal distributions, which is a prerequisite for further data analysis. A questionnaire with a coefficient of 0.7 or higher could be deemed as a questionnaire with satisfactory reliability (Nunnally 1994). The measurement model showcases acceptable reliability, as indicated by Cronbach's alpha coefficients reported before in the research instrument part, ranging from 0.845 to 0.899. Hair et al. (2010) suggest that factor loading, demonstrating the connection between items and factors, should have a value greater than ± 0.50 . The standardized factor loadings ranged from 0.55 to 0.78, which all fell within the acceptable range. Furthermore, Table 1 presents the assessment results of the validity. The validity of this model was checked using average variance extracted (AVE) as a pivotal criterion. It indicates the average shared variance between an index and a variable, with stronger validity scaling with the correlation between the variable and its index. The AVE's critical value is 0.4. The AVE values greater than 0.4 are deemed acceptable (Fornell and Larcker 1981). Composite reliability (CR) should be equal to or greater than 0.6. CR is a reliability index that is a measure utilized to evaluate the internal consistency of a series of indicators applied in the psychometrics and statistics realm. It is less biased than Cronbach's alpha (Hair et al. 2010). Additionally, the CR and AVE values spanned between 0.660 and 0.911 and from 0.400 to 0.524, respectively, signaling acceptable convergent validity. Moreover, Table 2 illustrates that the square root values of AVE of each construct were greater than the correlation coefficient values among the other constructs, signalling that the discriminant validity was acceptable.

4.3. Structural model

The goodness-of-fit indices and path coefficients were utilized to assess the SEM model. The hypotheses pertinent to the mediating role of needs satisfaction and creative self-

Table 1. Assessment of validity.

Latent variable	Dimension	SC	CR	AVE
Generative AI literacy in digital multimodal composing (GAILDMC)	AL	0.69–0.75	0.767	0.524
	BL	0.62–0.70	0.706	0.445
	CL	0.66–0.70	0.717	0.458
	EL	0.63–0.68	0.695	0.432
Needs Satisfaction (NS)	PA	0.67–0.70	0.724	0.467
	PC	0.62–0.65	0.672	0.406
	PR	0.65–0.74	0.746	0.496
	CISE	0.58–0.68	0.676	0.411
Creative Self-concept (CSC)	CPSE	0.71–0.72	0.757	0.509
	CPI	0.66–0.75	0.754	0.506
	CS	0.59–0.68	0.660	0.400
Self-regulated Learning of Writing (SRLW)	MS	0.64–0.72	0.721	0.463
	SBS	0.62–0.76	0.740	0.489
	MRS	0.65–0.77	0.754	0.507
		0.62–0.75	0.911	0.462
Generative AI literacy in digital multimodal composing (GAILDMC)		0.63–0.73	0.882	0.455
Needs Satisfaction (NS)		0.61–0.75	0.891	0.478
Creative Self-concept (CSC)		0.55–0.78	0.882	0.457
Self-regulated Learning of Writing (SRLW)				

Note. AL = affective learning; BL = behavior learning; CL = cognitive learning; EL = ethical learning; PA = perceived autonomy; PC = perceived competence; PR = perceived relatedness; CISE = creative ideas self-efficacy; CPSE creative performance self-efficacy; CSI = creative personal identity; CS = cognitive strategies; MS = metacognitive strategies; SBS = social behavior strategies; MRS = motivational regulation strategies. SC = standardized coefficients; CR = composite reliability; AVE = average variance extracted.

Table 2. Discriminant validity of potential variables.

Potential variable	GAILDMC	NS	CSC	SRLW
GAILDMC	0.679			
NS	0.597	0.674		
CSC	0.541	0.613	0.691	
SRLW	0.658	0.599	0.663	0.676

concept were tested using a bootstrapping process based on 5000 bias-corrected samples, a 95% bias-corrected confidence interval, and a 95% percentile confidence interval (Hayes and Scharkow 2013). This procedure enabled researchers to achieve robust estimations unaffected by the lack of multivariate normality (Hair et al. 2019). By imitating the data collecting process, bootstrapping is a reliable resampling strategy that solves possible restrictions through resampling the original data with replacement, generating new datasets. On each of these resampled datasets, the model is then re-estimated. Through the use of an iterative method, the indirect effects' sampling distribution may be enhanced, leading to the production of more accurate estimates of their standard errors and confidence intervals (Hair et al. 2010). All models were estimated using IBM AMOS version 28.0. Model quality of fitness of this study was evaluated using the Chi-Square (χ^2/df), Comparative Fit Index (CFI), Goodness of Fit Index (GFI), Parsimony Goodness-of-Fit Index (PGFI), Parsimony Comparative Fit Index (PCFI), Root Mean Square Error of Approximation (RMSEA), and Standardized Root Mean Square Residual (SRMR). The model was considered to have acceptable or good fit, respectively, if χ^2/df was less than 5 or 2 (Wheaton 1987), CFI was higher than 0.8 and 0.9 (Bentler 1990), GFI was higher than 0.8 or 0.9 (Escalera-Chávez, García-Santillán, and Venegas-Martínez 2014), PCFI and PGFI were higher than 0.6 or 0.8 (Blunch 2008), and RMSEA and SRMR were lower than 0.08 or 0.05 (Schermelleh-Engel, Moosbrugger, and Müller 2003; Yucel et al. 2020). As Table 3 displays, each value satisfied the suggested thresholds, implying that the structural model fits the data well.

4.4. Test of hypotheses

As depicted in Table 4 and Figure 2, the results revealed a positive and significant association between GenAI literacy in DMC and writing SRL ($\beta = 0.426, p < 0.001$), corroborating H1. A positive and significant correlation was also observed between GenAI literacy in DMC and needs satisfaction ($\beta = 0.809, p < 0.001$), supporting H2. There is a positive relationship between needs satisfaction and writing SRL. However, this relationship seems insignificant ($\beta = 0.063, p = 0.562$), which proves that needs satisfaction is not significantly positively correlated with writing SRL, rejecting H3. Since H4 can be supported if both H2 and H3 are correct, it can be concluded that needs satisfaction insignificantly

Table 3. Goodness-of-fit indicators for the structural model.

Fit index	Good criteria	Acceptable criteria	Value of SEM model
CMIN/DF (χ^2/df)	< 2	< 5	3.655
Goodness of Fit Index (GFI)	> 0.9	> 0.8	0.818
Parsimony Goodness-of-Fit Index (PGFI)	> 0.8	> 0.6	0.717
Comparative Fit Index (CFI)	> 0.9	> 0.8	0.854
Parsimony Comparative Fit Index (PCFI)	> 0.8	> 0.6	0.787
Root mean square error of approximation (RMSEA)	< 0.05	< 0.08	0.065
Standardized Root Mean Square Residual (SRMR)	< 0.05	< 0.08	0.049

Table 4. The test results of path relationship.

Hypothesis	Path	Unstand estimates	t	Sig.	Stand estimates	Hypothesis test
H1	GAILDMC->SRLW	0.460	6.950	***	0.426	Supported
H2	GAILDMC->NS	0.897	13.070	***	0.809	Supported
H3	NS->SRLW	0.061	0.579	0.562	0.063	Rejected
H5	GAILDMC->CSC	0.237	3.942	***	0.221	Supported
H6	CSC->SRLW	0.569	4.463	***	0.565	Supported
H8	NS->CSC	0.720	10.057	***	0.745	Supported

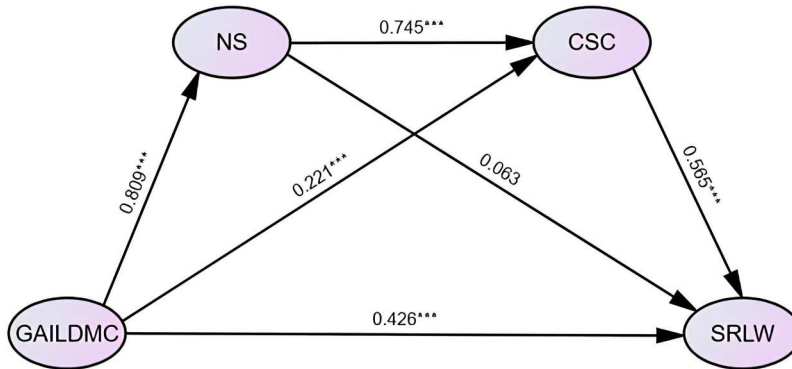


Figure 2. The path diagram.

mediates the association between GenAI literacy in DMC and Chinese EFL learners' writing SRL, rejecting H4. GenAI literacy in DMC has been discovered to have significant correlations with creative self-concept ($\beta = 0.221$, $p < 0.001$), justifying H5. Creative self-concept was found to be significantly positively linked to writing SRL ($\beta = 0.565$, $p < 0.001$), substantiating H6. H7 can be supported if both H5 and H6 are valid. Therefore, it could be implied that creative self-concept significantly mediates the association between GenAI literacy in DMC and writing SRL, evidencing H7. The two mediation factors, needs satisfaction and creative self-concept, were significantly linked to each other ($\beta = 0.745$, $p < 0.001$), lending credence to H8. To summarise, the empirical data supported the hypotheses of H1, H2, H5, H6, H7, H8, and H9 while rejecting H3 and H4.

4.5. Analyses of the mediation roles of needs satisfaction and creative self-concept

To respond to RQ1, the results presented in Table 5 show that the direct and total prediction effects of GenAI literacy in DMC on EFL learners' writing SRL were significant. The total effect of EFL learners' GenAI literacy in DMC on writing SRL was 1.017, ($t = 10.594$, 95% confidence interval (bias-corrected) [0.833, 1.214], $p < 0.001$), and the direct effect was 0.460, ($t = 4.510$, 95% confidence interval (bias-corrected) [0.273, 0.673], $p < 0.001$). To answer RQ2, the path of GAILDMC->NS->SRLW indicated the nonsignificant indirect effect of 0.055, ($t = 0.055$, 95% confidence interval (bias-corrected) [-0.291, 0.298], $p = 0.674$). As indicated in Table 5, the indirect effect of needs satisfaction accounts for 9.9% of the overall indirect effect. This result denotes that needs satisfaction didn't mediate the relationship between GenAI literacy in DMC and writing SRL. To address

Table 5. Total, direct, and indirect effects of the theoretical model.

Paths	Point	Product of		Bootstrapping			
	Estimate	Coefficient		Bias-corrected 95% CI		Percentile 95% CI	
		SE	t value	Lower	Upper	Lower	Upper
Evaluation of direct, indirect, total effects, and comparison of indirect effects							
(H9) GAILDMC→NS→CSC→SRLW	0.367	0.133	2.759	0.178	0.706	0.176	0.695
(H4) GAILDMC→NS→SRLW	0.055	0.148	0.372	−0.291	0.298	−0.299	0.294
(H7) GAILDMC→CSC→SRLW	0.135	0.060	2.250	0.047	0.296	0.035	0.267
TIE	0.557	0.076	7.329	0.424	0.729	0.415	0.714
DE	0.460	0.102	4.510	0.273	0.673	0.273	0.673
TE	1.017	0.096	10.594	0.833	1.214	0.835	1.214
DistalIE-NSIE	0.312	0.272	1.147	−0.100	1.001	−0.106	0.983
CSCIE-NSIE	0.080	0.181	0.442	−0.233	0.488	−0.226	0.498
DistalIE-CSCIE	0.232	0.130	1.785	0.068	0.597	0.062	0.556
Percentage of indirect effects							
DistalIE/TIE	0.659	0.244	2.701	0.331	1.295	0.326	1.275
NSIE/TIE	0.099	0.270	0.367	−0.549	0.529	−0.557	0.526
CSCIE/TIE	0.242	0.095	2.547	0.083	0.459	0.068	0.439
TIE/TE	0.548	0.076	7.211	0.404	0.706	0.402	0.704
Percentage of direct effects							
DE/TE	0.452	0.076	5.947	0.294	0.596	0.296	0.598

Note. IE = indirect effect; TIE = total indirect effect; DE = direct effect; TE = total effect; DIE = distal indirect effect.

RQ3, the path of GAILDMC→CSC→SRLW denotes a significant indirect impact of 0.135 ($t = 2.250$, 95% confidence interval (bias-corrected) [0.047, 0.296], $p = 0.004$). The indirect effect of creative self-concept accounts for 24.2% of the total indirect effect. This research finding evidenced that creative self-concept didn't mediate the relationship between GenAI literacy in DMC and writing SRL. To resolve RQ4, the path of GAILDMC→NS→CSC→SRLW revealed a significant indirect effect of 0.367 ($t = 2.759$, 95% confidence interval (bias-corrected) [0.178, 0.706], $p < 0.001$). The distal indirect effect of needs satisfaction and creative self-concept accounts for 65.9% of the total indirect effect. The total indirect effects take up 54.8% of the total effects. This research discovery suggested that creative self-concept and needs satisfaction serially and significantly mediated the relationship between GenAI literacy in DMC and writing SRL.

We have also conducted comparisons between these different indirect effects, DistalIE (GAILDMC→NS→CSC→SRLW) has significantly larger effects than CSCIE (GAILDMC→CSC→SRLW), with a comparison effect of 0.232 ($t = 1.785$, 95% confidence interval (bias-corrected) [0.068, 0.597], $p = 0.001$). There are no significant differences between DistalIE and NSIE ($t = 1.147$, 95% confidence interval (bias-corrected) [−0.100, 1.001], $p = 0.143$), or CSCIE and NSIE ($t = 0.442$, 95% confidence interval (bias-corrected) [−0.233, 0.488], $p = 0.644$).

5. General discussion and implications

5.1. Empirical contributions

In short, our study demonstrated that creative self-concept served as a significant predictor in the relationship between GenAI literacy in DMC and writing SRL, while needs satisfaction did not. Furthermore, the chain mediation effects of needs satisfaction and creative self-concept were the most significant compared with other mediation effects. To resolve RQ1, our results validated that GenAI literacy in DMC had a direct prediction effect on writing SRL, which was consistent with that of prior research claiming that the integration

of GenAI could enhance each phase of SRL (forethought, performance, and self-reflection) in writing by providing immediate feedback and individualized learning materials, thus prompting reflection on writing SRL strategies and outcomes (Chang et al. 2023; Lai 2024). The study also corroborated Abdelhalim's (2024) study, which assumed that DMC would create a supportive and compatible student-centred learning environment for exercising writing SRL strategy use skills.

To address RQ2, our study identified that needs satisfaction was not a significant predictor of writing SRL in its proposed mediation effects between GenAI literacy in DMC and writing SRL, which diverged from past research (Xia, Chiu, and Chai 2023a; Xia et al. 2023b) that corroborated the significant mediation effects of needs satisfaction. According to the trade-off hypothesis (Skehan 2009), cognitively demanding tasks necessitate trade-offs due to limited attentional mechanisms. The tertiary EFL learners in our study are more inclined to utilise the GenAI tools for more complex academic or argumentative writing tasks, which were unlike narrative writing tasks, such as storytelling conducted in the secondary school or primary school classrooms. These university EFL learners may prioritise performance strategies such as prompt design and task efficiency over seeking satisfaction of their internal psychological needs (Kim et al. 2025). These writing assistance objectives were in stark contrast with the young writers from middle school or elementary schools, who attached more importance to the satisfaction of needs for satisfying the intrinsic motivations, such as enjoyment or interaction pleasure, when interacting with GenAI platforms to assist their writing performances (Makransky et al. 2025).

These differences could also be attributed to the compensation theory from cognitive psychology (Ackerman 1988), as working memory could be less taxed for those university writers with higher level of domain-specific GenAI-assisted writing knowledge. Thus, these writers attached more importance to the external performance than the intrinsic needs satisfaction when using GenAI for writing assistance, which is quite different from those young learners who prioritized the basic needs satisfaction since they were at the developmental phase where their cognitive load is lower than those older learners (Ackerman 1988; Vasylets and Marín 2021). Furthermore, sociocultural context may also be taken into account, as educational practitioners usually conduct enjoyment-directed GenAI-assisted writing instruction in EFL classrooms, such as executing gamification project-based learning for young learners (Fitriyani et al. 2025; Ng, Tan, and Leung 2024), while other educators tend to carry out utility-directed instructional activities, such as implementing feedback awareness-enhancing activities for university EFL learners (Shi et al. 2025; Tam 2025). These cognitive, developmental, and social-cultural factors collectively resulted in the unexpected result that needs satisfaction failed to significantly mediate the relationship between GenAI literacy and writing SRL.

To respond to RQ3, GenAI literacy in DMC indirectly predicted writing SRL, with creative self-concept operating as a significant mediator. This finding echoed previous studies, which claimed that students' GenAI literacy in DMC shaped creative possibilities for expression (Jiang 2024; Smith et al. 2025). The reason may lie in that creative self-concept may exert pivotal influences on both students' unconscious and conscious writing SRL processes. Creative self-concept could stimulate students to approach assignments from various points of view, thus activating the most relevant schema for unconscious writing SRL (Wang and Lajoie 2023). For conscious SRL, creative related strategies such as mind-mapping could alleviate the cognitive loads in writing SRL processes (Fang et al. 2023).

To address RQ4, GenAI literacy in DMC was found to indirectly predict writing SRL through a serial mediation pathway involving needs satisfaction and creative self-concept. The findings of significant chain mediation effects of needs satisfaction and creative self-concept were consistent with prior research (Lin et al. 2025; Nong, Wei, and Li 2024). While needs satisfaction alone did not significantly mediate the relationship between GenAI literacy in DMC and writing SRL, it could significantly predict writing SRL through creative self-concept. This finding proved that the prediction influences of needs satisfaction on writing SRL are possibly indirect and may hinge on the activation of creative self-concept, rather than in a direct way. This finding was congruent with previous scholars' claim that need-supportive environments were likely to induce more creative involvement because they generated a greater sense of freedom and safety, which may incur more creative activities for the writing SRL process (Highland and Fedtke 2023; Wünsch-Nagy 2025; Zhang and Yu 2025). Creative self-concept helps translate the innate motivation into concrete actions of generating valuable and novel ideas during the problem-solving process in the writing SRL process (Cromwell, Haase, and Vladova 2023). Therefore, needs satisfaction alone may not directly predict SRL behaviors in writing – unless there's an additional mechanism that translates that motivation into strategic writing SRL behaviors.

Moreover, the cultural and educational factors may have some impacts on our research findings. Chinese college students had a preference to use GenAI tools as language assistants instead of a content generator in their EFL writing SRL process (Yang, Rau, and Ma 2025). And the translation of Chinese text into English was the most frequently used mode among Chinese EFL learners. In GenAI-assisted DMC processes such as text-to-video resemiotisation, Chinese EFL learners were more inclined to engage in creating and revising search queries to translate and visualise their ideas for the video format (Li, Xia, and Guo 2025). This was different from other EFL learners from U.S. and Japan who exhibited rather limited interest in using GenAI for translating from their native language to English during writing or multimodal writing SRL process (Scherr et al. 2025). That is to say, in the Chinese EFL context, it is not the satisfaction of autonomous translation practices that leads to writing SRL but rather the creative expression ways shaped by students' creative self-concept in shaping and refining prompts that externalize the autonomous translation and translanguaging practices into visible SRL writing behaviors. This specific cultural context of our study further explicates the results of our study that needs satisfaction could significantly mediate the relationship between GenAI literacy in DMC and writing SRL only through coupling with the creative self-concept.

Furthermore, compared with other Chinese EFL writers with relatively low English writing proficiency from primary schools of middle schools who used GenAI for expansion and support in their SRL drafting stage as they depend on their first language for idea generation and understanding, Chinese university EFL learners in our study were inclined to utilise GenAI for strategic enhancement and expediency throughout the outlining, drafting, and note-taking SRL writing or DMC phases (García and Kano 2014; Yang and Lin 2025; Zheng and Drybrough 2023). Chinese university learners' tendency to utilise GenAI tools for expediency and strategy improvement suggests they rely more on the creativity of the prompts to obtain the desired scaffolding results than on the satisfaction of basic psychological needs. In addition, as academic writing is a relatively demanding task of high cognitive requirements (Liu et al. 2024; 2025; Jiang and Lai 2025; Xie and

Jiang 2025), students need to generate novel ideas and content throughout the whole process of writing, therefore relying more on their creative self-concept to get these needs satisfied through GenAI for assisting their writings.

5.2. Theoretical implications

Our study has some insightful theoretical implications. It was found that needs satisfaction did not directly predict writing SRL, but instead exerted an indirect impact on writing SRL through the mediator of creativity self-concept, which was in contrast with the core hypothesis surrounding the outcomes of SDT theory from prior relevant research supporting the existence of a direct relationship between EFL learners' basic psychological needs (autonomy, competence, and relatedness) and learners' writing SRL behaviors (Wang, Wang, and Xu 2025; Xu and Zhou 2024). This research discovery challenges these theoretical claims by incorporating creative self-concept as a former underexplored mediational motivational-cognitive factor between needs satisfaction and writing SRL (Li, Li, and Chen 2018). Learners' creative self-concept led them to transfer the SRL skills in GenAI-assisted DMC to a monomodal writing task. The satisfaction of EFL learners' needs is not yet sufficient since the SRL writing skill transfer needs to be activated through learners' creative agency across different task modalities (Cimasko and Shin 2017; Wu and Albert 2024). This creative transfer suggests that writing SRL is not merely triggered by needs satisfaction, but co-constructed through learners' perceptions of their own creative agency across task modalities, which makes creativity a critical connection between needs satisfaction and writing SRL.

Furthermore, our research discoveries extended the Metacognition Self-regulated Learning acceptance (MSRL) model in the GenAI context (Dahri et al. 2024). In the MSRL model, concepts such as perceived usefulness, trust, enjoyment, and personal competence, which symbolise the internal psychological states, are postulated to impact SRL through the mediation effects of learners' attitudes towards utilizing ChatGPT. Our study enriched this theoretical model by revealing that EFL learners' perceived competence in GenAI in DMC could directly influence their writing SRL. Furthermore, there also existed an indirect dual mediational path containing needs satisfaction of these psychological constructs and creative self-concept. Therefore, the future MSRL model should be extended to involve the direct influence of students' GenAI literacy in DMC on writing SRL, that is, to stress the salient importance of cognitive self-efficacy beliefs of these students' capability to utilise GenAI for their DMC to serve as strong motivational support to help EFL learners initiate, maintain, and devote more learning efforts to utilise writing SRL strategies (Guan et al. 2025; Ji et al. 2025; Teng 2021). This model should also be adapted and expanded to involve the creative self-concept as an essential cognitive-motivational mediator to work together with these internal psychological needs mechanism factors to affect students' writing SRL behaviors (Papachristopoulos et al. 2023; Weng et al. 2025).

5.3. Practical implications

In terms of practical implications, our research may also contribute to the further curriculum syllabus that integrates GenAI literacy in DMC development as an entry point to promote EFL learners' writing SRL.

To begin, our findings advocate for the integration of GenAI literacy into EFL curriculum could scaffold Chinese EFL learners' writing SRL. Future practitioners should put equal emphasis on fostering students' DMC practices by instilling visual grammar, introducing a storyboard, encouraging multimodal critical thinking, and enhancing students' GenAI technological skills by training them on how to craft effective prompts. The storyboarding and visual grammar could be effective for students' planning and reflecting during the writing SRL process. The visual grammar might help students understand the intermodal relationship more deeply through multimodal discourse analysis of the DMC samples (Huang and Xia 2024; Ma and Jiang 2025). The structured storyboard could provide the EFL learners with organized and clear guidance for planning (Balzotti 2016; Kim et al. 2022).

Secondly, further pedagogical practices may consider incorporating students' self-made multimodal GenAI-based retrieval augmented generation (RAG) chatbots in DMC classrooms to exercise EFL learners' GenAI literacy in DMC to promote their writing SRL. Through self-made multimodal GenAI-based RAG chatbots, students can personalize the multimodal GenAI functions to meet their psychological needs for regulating their writing behaviors. For instance, by setting the relevant parameters such as temperature in the chatbot, these multimodal GenAI chatbots may generate diversified or coherent responses to satisfy students' autonomy needs for regulating their SRL drafting process in their writing (Guo and Li 2024; Wu et al. 2024).

Last but not least, future practitioners should bear in mind to foster an autonomous and creative environment to adapt to each EFL learner's literacy skills by applying learning dashboard analytics in the GenAI-assisted writing classrooms. Incorporation of learning dashboard analytics can help educators monitor how EFL learners harness their GenAI literacy in DMC to satisfy their own needs and induce creative self-concept. In this way, educators could design a suitable curriculum to help EFL learners transfer the self-regulation skills fostered in the GenAI-assisted DMC environment to their monomodal writing context.

6. Limitations and future study

Several limitations of this study warrant being pointed out. First, the cross-sectional design of this study may need a stronger justification for delving into the causal relationships between GenAI literacy in DMC, needs satisfaction, creative self-concept, and writing SRL. Although significant relationships were identified, a longitudinal study such as latent profile transition analysis would be more appropriate to track the progression of these variables over time, establishing a more robust causality between these variables (Jeong and Feldon 2023). Secondly, the use of self-reported measures (i.e. Likert-scale questionnaires) to assess students' GenAI literacy in DMC, creative self-concept, needs satisfaction, and SRL in writing implies inherent limitations, as these subjective reports may not reflect EFL writers' actual performance or observable behavior. Therefore, future research needs to complement our research findings with objective assessment methods such as peer assessment and teacher evaluations, writing samples analysed for SRL strategies, teacher ratings, or log data from GenAI tools to offer a more holistic, comprehensive, and validated view of students' writing SRL (Tian, Liu, and Zhang 2022).

Third, the sample gathered in this study is demographically constrained to specific cultural and age backgrounds, limiting the generalisability of the findings. Participants in this

study were 58.5% Chinese undergraduate students, and generally representing adults with a mean age of 24.67. The cultural and age characteristics of the sample may influence students' familiarity with GenAI tools or their writing SRL practices, which would limit the applicability of these research results to younger or non-Chinese populations. For instance, some scholars have found that older users tended to be more familiar with GenAI tools as they were more engaged in more function-oriented interactions and were more familiar with GenAI applications than the younger users (Sundar 2020). Furthermore, college-level students were found to demonstrate more SRL abilities during interactions with the GenAI chatbots, especially with regard to self-evaluation and using task strategies (Xu et al. 2025), when compared with secondary students who tended to display more environmental structuring and goal setting writing SRL strategies in GenAI settings (Ng, Tan, and Leung 2024). Therefore, more studies warrant to be carried out to find out whether the results of our study could be generalised to other ages or cultural groups.

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CRediT author statement

Liu: conceptualisation; data collection; formal analysis; writing-original draft;

Zhang: conceptualisation; methodology; investigation; writing; review, revising & editing;

Neufeld: review, revising & editing.

Human ethics approval

The study has been reviewed and given approval by The University of Auckland Committee on Human Participation (Human Ethics Approval number: UAHPEC28436).

Data availability statement

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

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